Korea Technology and Information Promotion Agency for SMEs

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# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | Spatial Omics & Translational Modeling Platform for Target-Specific Therapy and Immunotherapy Response Prediction in Head and Neck Cancer | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *Biotechnology / Biohealth* | | | |
| | Medium Category | Small Category | Medium Category | Small Category |
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| Concept of Proposed Technology | This proposal introduces an institute-led translational research framework that integrates:<br>• **Clinically annotated archival head and neck squamous cell carcinoma (HNSCC) tissues**<br>• **Spatial transcriptomics (& expandable spatial multi-omics)**<br>• **Existing multi-omics datasets (genomics/transcriptomics and/or available molecular profiles)**<br>• **Longitudinal clinical history and treatment outcomes**<br>The core concept is to generate **spatially resolved tumor and microenvironment maps** and develop **AI-driven modeling frameworks** to enable:<br>1. **Target prioritization and evaluation** for **target-specific modalities**, including antibody-drug conjugates (**ADCs), radiopharmaceutical therapies (RPT), and T cell engagers (TCE)** | | | |

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| | 2. **Predictive modeling of immunotherapy response and resistance** based on spatial immune phenotypes, resistance niches, and multi-modal clinical integration<br><br>This technology aims to produce **clinically actionable biomarkers, target packages, and modeling outputs** that can directly inform drug development programs, patient selection, and trial enrichment strategies. A key objective is to develop a standardized and reusable workflow that can be applied beyond HNSCC to broader range of cancer types. |
| **Rationale for Joint R&D Project** | Advanced HNSCC remains a highly lethal disease with limited therapeutic options, no novel therapeutics introduced in the past decade, and a median survival of less than 14 months. While immune-checkpoint inhibitors (ICI) have transformed treatment paradigm, clinical benefit is restricted to a minority of patients, and no validated, reproducible biomarkers exist to guide patient selection. These gaps represent a major unmet medical and industrial need, underscoring the urgency for innovative drug-development strategies and clinically deployable biomarkers to enable precision immuno-oncology in advanced HNSCC.<br><br>HNSCC exhibits substantial **inter- and intra-tumoral heterogeneity**, and many promising therapeutic programs fail due to:<br><br>• insufficient **tumor selectivity**<br>• **spatially heterogeneous target expression** causing incomplete response and resistance<br>• **microenvironment-driven resistance niches**, including immune exclusion and stromal barriers<br>• limited predictive power of biomarkers for treatment response<br><br>Joint R&D is required because successful translation demands a combined capability set:<br><br>• the institute's strengths in **clinically annotated cohorts, longitudinal outcomes, and translational oncology expertise**<br>• and the industry partner's capability in **scalable spatial profiling, advanced analytics, and platform-driven biomarker development**<br><br>Through this collaboration, Korean SMEs can co-develop globally competitive spatial biomarker and target discovery solutions, accelerate drug development readiness, and generate partnership-grade deliverables aligned with future pharma collaboration opportunities |
| **Key Words** | 1. **Spatial Transcriptomics**<br>2. **Head and Neck Cancer**<br>3. **Tumor Microenvironment**<br>4. **Treatment Response Prediction**<br>5. **Translational Multi-Omics Modeling** |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level ( 2 ) | End of the Project | Level ( 3 ) |
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| R&D Objectives | The R&D project aims to develop and validate a spatially informed translational R&D framework for HNSCC that supports both target-specific drug development and immunotherapy response prediction, enabling downstream clinical and industry partnership applications. | | | |
| R&D Target Specifications | Specific objectives include:<br>1. **To develop a spatially validated target prioritization for target-specific modalities**<br>2. **To define spatial immune phenotype and create resistance niche mapping for ICI-treated cohorts**<br>3. **To construct integrated predictive modeling frameworks**<br>4. **To generate translational deliverables for industry partnerships and joint publications** | | | |
| R&D Plan | **Phase 1: Setup & pilot feasibility**<br>• Clinical and translational sudy design, sample selection, and workflow/QC setup<br>• Pilot spatial profiling and initial spatial feature generation<br>• Data harmonization for clinical, spatial, and multi-omics integration<br>**Phase 2: Scale-up & target and TME landscape mapping**<br>• Scaling of spatial profiling across the discovery cohort<br>• Quantification of tumor-selective expression and heterogeneity of candidate targets<br>• Generation of initial modality-aligned target prioritization methods<br>**Phase 3: Tumor immunemicroenvironment modeling**<br>• Spatial immune phenotyping of ICI-(or other immune-related) treated cohorts<br>• Identification of resistance niches and predictive spatial biomarkers<br>• Development of response/resistance prediction models<br>**Phase 4: Validation & translational packaging**<br>• Consolidatation and validation of therapeutic targets and predictive biomarkers<br>• Finalization of reproducible analytical pipelines<br>• Preparation of outputs supporting publications and downstream industry partnering | | | |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | The research institute will provide the clinical and translational foundation required for a high-impact spatial biology program in HNSCC, including:<br><br>• **Clinical resources and study scope**<br>  ○ Provide clinically annotated archival HNSCC tissues and associated metadata<br>  ○ Provide longitudinal clinical history, including treatment exposure and outcomes (response, durability, survival), where available<br>  ○ Define clinically meaningful study endpoints and translational use-cases<br>• **Scientific leadership and translational direction**<br>  ○ Lead study design, cohort definition, sample selection strategy, and hypothesis framing<br>  ○ Interpret results in clinical context and ensure outputs are aligned with decision-making needs<br>• **Multi-omics integration support**<br>  ○ Provide existing multi-omics datasets (as available) and support integrative analysis strategies<br>  ○ Enable translational validation planning (e.g., orthogonal evidence and clinically relevant checks)<br>• **Collaboration and dissemination**<br>  ○ Support iterative scientific discussions, milestone reviews, and joint reporting<br>  ○ Contribute to joint publications, scientific presentations, and downstream partnering materials, as appropriate |
| **Expected R&R of Korean SME Partner** | • **Spatial profiling and technical execution**<br>  ○ Provide a robust spatial transcriptomics platform suitable for archival tissues<br>  ○ Execute sample processing, assay workflows, and quality control with documented metrics<br>  ○ Generate standardized spatial data outputs and analytical-ready datasets<br>• **Data engineering and computational development**<br>  ○ Develop analysis pipelines for tissue compartment annotation (tumor/stroma/immune) and spatial feature extraction<br>  ○ Build integrative modeling workflows combining spatial features + clinical variables + molecular profiles<br>  ○ Deliver interpretable outputs<br>• **Modeling, validation, and deliverable packaging**<br>  ○ Develop modality-aligned scoring frameworks for target evaluation (selectivity, heterogeneity, risk flags)<br>  ○ Train and validate predictive models for treatment response/resistance, including robustness assessment<br>• **Collaboration commitment**<br>  ○ Participate in regular technical reviews and milestone-based progress reporting<br>  ○ Support co-authorship, joint dissemination, and technology transfer/partnership discussions where relevant |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | This joint R&D project is expected to deliver a spatially informed translational framework in HNSCC that enables both target-specific therapy development and immunotherapy response prediction. Key outcomes include a prioritized portfolio of spatially validated therapeutic targets, interpretable spatial biomarkers, and integrated predictive models linked to real-world clinical outcomes.<br>The impact will be improved drug candidate prioritization, more optimized patient selection and trial enrichment strategies, and accelerated industry partnership readiness, supporting Korean SMEs in building globally competitive spatial biomarker and precision oncology capabilities. |